

Magnetic Fields Due to Currents

Chapter 29-1 R.H.W 10th Edn

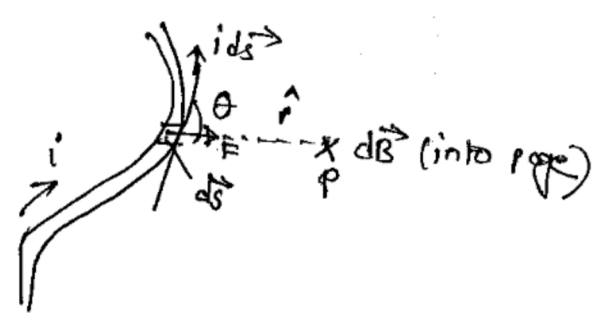
Magnetic Fields Due to Currents

Biot - Savart law Statement: This law asserts that the

contribution $d\vec{B}$ to the field produced by a current-length element $i d\vec{s}$ at a point P located a distance r from the current element is

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{s} \times \hat{r}}{r^2} \quad (\text{Biot-Savart law})$$

⊙ We wish to calculate the net field $d\vec{B}$ produced at P by a typical current-length element.



⊙ From experiment we find that magnetic fields, like electric fields, can be superimposed to find a net field.

⊙ A charge element dq producing an electric field is scalar, a current-length element $i d\vec{s}$ producing a magnetic field is vector, now fields are present in scalar and in vector.

Magnitude The magnitude of the field $d\vec{B}$ produced at point P at distance r by a current-length element $i d\vec{s}$ turns out to be

$$dB = \frac{\mu_0}{4\pi} \frac{i ds \sin\theta}{r^2}, \quad \text{where } \theta \text{ is the angle between the direction of } d\vec{s} \text{ and } \vec{r} \text{ (} d\vec{s} \text{ towards P), } \mu_0 \text{ is a constant, called the permeability constant, } \mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A} \approx 1.26 \times 10^{-6} \text{ T}\cdot\text{m/A}.$$

Direction The direction of $d\vec{B}$ is the cross product $d\vec{s} \times \hat{r}$.

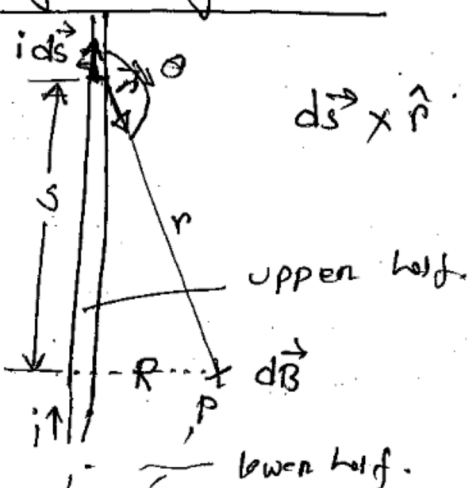
$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{s} \times \hat{r}}{r^2} \quad (\text{Biot-Savart law})$$

This vector equation and its scalar form, are known as the law of Biot and Savart. The law, experimentally deduced to an (inverse-square) law.

Note: We use this law to calculate the net magnetic field $\vec{B}$ produced at a point by various distribution of current.

Magnetic field Due to a Current in a Long Straight wire

We seek the field $\vec{B}$ at point P , a perpendicular distance R from the wire. The magnitude of the differential magnetic field produced at P by the current-length element $id\vec{s}$ located a distance r from P is given by



$$dB = \frac{\mu_0}{4\pi} \frac{id\vec{s} \sin\theta}{r^2} \quad | \quad d\vec{B} \text{ at point } P \text{ has}$$

The magnetic field produced by the lower half of the wire is exactly the same as that produced by the upper half.

the same direction for all the current-length elements into which the wire can be divided

$$\text{Total } \vec{B} = 2 \int_0^R dB = \frac{\mu_0 i}{2\pi} \int_0^\pi \frac{\sin\theta ds}{r^2}$$

$$B = \frac{\mu_0 i}{2\pi} \int_0^\pi \frac{\sin\theta ds}{(s^2 + R^2)^{3/2}}$$

$$= \frac{\mu_0 i}{2\pi R} \left[\frac{s}{(s^2 + R^2)^{1/2}} \right]_0^\pi = \frac{\mu_0 i}{2\pi R}$$

for semi infinite wire B is half of the current B

θ, s and r in this equation are not independent

$$r = \sqrt{s^2 + R^2}$$

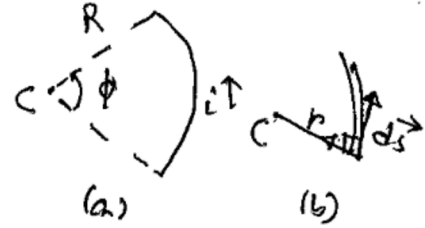
$$\sin\theta = \sin(\pi - \theta)$$

$$= \frac{R}{\sqrt{s^2 + R^2}}$$

Magnetic Field Due to a Current in a Circular Arc of Wire.

(3)

(a) Arc-shaped wire with central angle ϕ , radius R , and center C , carrying current i .



At C , each current-length element $i ds$ of the wire produces a magnetic field of magnitude dB

(b) vectors ds and $\hat{r}$ is 90° ; also $r = R$. Thus by substituting R for r and 90° for θ , from previous equation we obtain $\rightarrow$

$$dB = \frac{\mu_0}{4\pi} \frac{i ds \sin 90^\circ}{R^2} = \frac{\mu_0}{4\pi} \frac{i ds}{R^2}$$

$$ds = R d\phi$$

$$B = \int dB = \int_0^\phi \frac{\mu_0}{4\pi} \frac{i R d\phi}{R^2}$$
$$= \frac{\mu_0 i}{4\pi R} \int_0^\phi d\phi$$

$$= \frac{\mu_0 i \phi}{4\pi R} \quad (\text{at the center of circular arc})$$

$$B = \frac{\mu_0 i (2\pi)}{4\pi R} = \frac{\mu_0 i}{2R} \quad (\text{at the center of full circle})$$

Ampere's Law

①

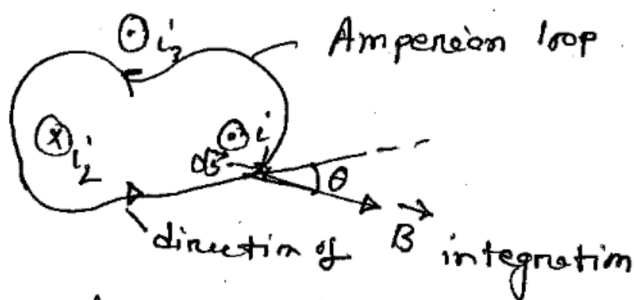
→ Recall, If the charge distribution is ~~complicated~~ has, planar, cylindrical or spherical symmetry, we can apply Gauss' law to find the net electric field with considerably less effort.

→ Similarly, we can find the net magnetic field due to any distribution of currents by first writing differential magnetic field $d\vec{B}$ due to a current-length element and then summing the contributions of $d\vec{B}$ from all elements.

→ However, if the distribution has some symmetry we may be able to apply Ampere's law to find the magnetic field with considerably less effort.

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} \quad (\text{Ampere's Law})$$

$\oint \vec{B} \cdot d\vec{s}$ is to be integrated around a close loop, called an Ampere's loop by that closed loop.
 net current i_{enc} encircled



$$\begin{aligned} \oint \vec{B} \cdot d\vec{s} &= \oint B \cos \theta ds = \mu_0 i_{enc} \\ &= \mu_0 (i_1 - i_2) \end{aligned}$$

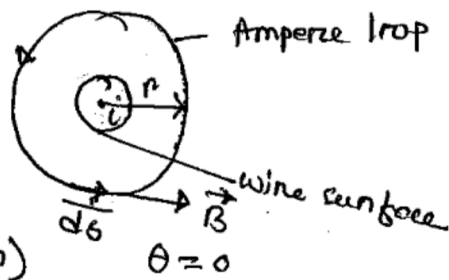
Magnetic Field Outside a long Straight wire with current

$$\oint \vec{B} \cdot d\vec{s} = \oint B \cos \theta ds$$

$$= B \oint ds = B(2\pi r)$$

$$\Rightarrow B 2\pi r = \mu_0 i$$

$$\boxed{B = \frac{\mu_0 i}{2\pi r}} \text{ outside straight wire.}$$



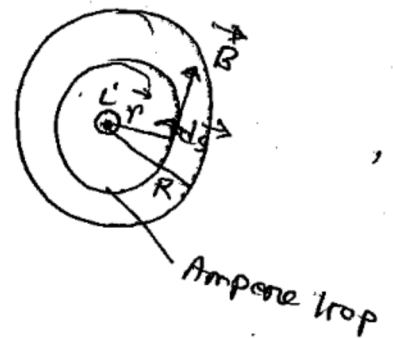
Magnetic Field Inside a long Straight wire with current

$$\oint \vec{B} \cdot d\vec{s} = B \oint ds = B(2\pi r)$$

$$I_{enc} = i \frac{\pi r^2}{\pi R^2}$$

$$B(2\pi r) = \mu_0 i \frac{\pi r^2}{\pi R^2}$$

$$\boxed{B = \left(\frac{\mu_0 i}{2\pi R^2} \right) r} \text{ [(inside straight wire)]}$$



⑧ A Current-Carrying Coil as a Magnetic Dipole

$$\vec{\tau} = \vec{\mu} \times \vec{B} \dots \dots$$

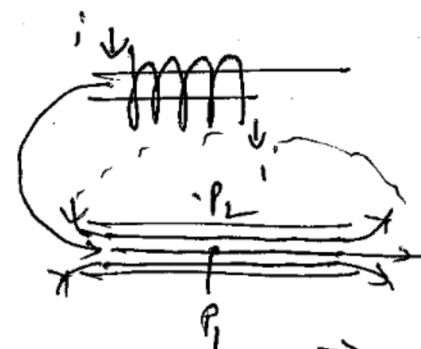
Here $\mu \rightarrow$ magnetic dipole moment of the coil. Have magnitude $NI A$. (N - number of turns), i is the current in each turn and A is the area enclosed by each turn.

Magnetic field of a coil

Apply Ampere's law in solenoids

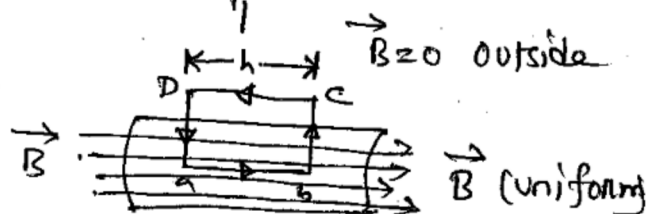
Ampere's law,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} \quad (1)$$



For rectangular Amperian loop
abcd. We write

$$\oint \vec{B} \cdot d\vec{s} = \int_a^b \vec{B} \cdot d\vec{s} + \int_b^c \vec{B} \cdot d\vec{s} + \int_c^d \vec{B} \cdot d\vec{s} + \int_d^a \vec{B} \cdot d\vec{s} \quad (2)$$



$$\oint \vec{B} \cdot d\vec{s} = Bh + 0(90^\circ) + 0(\text{outside of the solenoid}) + 0(90^\circ) \\ = Bh$$

net current $i_{enc} = inh$ $[n \rightarrow \text{number of turns per unit length}]$

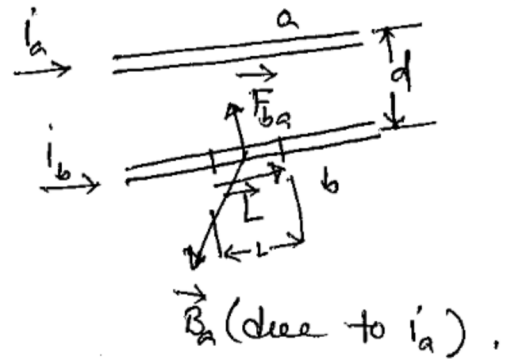
$$Bh = \mu_0 in h$$

$$B = \mu_0 in \text{ (ideal solenoid)}$$

Force Between Two parallel currents

The magnitude of $\vec{B}_a$ at every point of wire b is,

$$B_a = \frac{\mu_0 i_a}{2\pi d} \quad \text{--- (1)}$$



The force $\vec{F}_{ba}$ on a length L of wire b due to the external magnetic field $\vec{B}_a$ is

$$\vec{F}_{ba} = i_b \vec{L} \times \vec{B}_a \quad \left[\vec{L} \rightarrow \text{length vector of the wire} \right]$$

$$F_{ba} = i_b L B_a \sin 90^\circ = \frac{\mu_0 L i_a i_b}{2\pi d}$$

[F_{ba} direction $\vec{L} \times \vec{B}_a$]
directly towards wire a.

$$\boxed{F_{ba} = \frac{\mu_0 L i_a i_b}{2\pi d}}$$